QUESTION BANK 2025–27

Organised module-wise • All 126 original questions retained

MODULE 1 — DATA SCIENCE FUNDAMENTALS

1. Define Data Science. Explain the major steps involved in the Data Science lifecycle.

2. Explain the classification of Data Science based on its major areas or types. Give suitable examples.

3. List and explain any four applications of Data Science in real-world domains.

4. What is the Data Science Process? Explain its importance in solving real-world problems.

5. What is Prior Knowledge in Data Science? Explain how prior knowledge helps in understanding and analysing a dataset.

6. Explain Data Preparation and Modelling in the Data Science Process.

7. Define a dataset. Explain the different components of a dataset with a suitable example.

8. Explain the different types of data used in Data Science with suitable examples.

9. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

10. Explain the importance of descriptive statistics in summarizing univariate and multivariate data.

11. Explain the concept of descriptive statistics and describe how it can be summarized using suitable statistical measures.

12. Differentiate between Univariate Data and Multivariate Data with suitable examples.

13. Explain Scatter Plot, Bubble Chart, and Density Chart. Mention one application of each.

14. Explain a Histogram and a Quartile Plot with suitable examples.

15. Explain Scatter Plot, Bubble Chart, and Density Chart. Mention one application of each.

76. Explain Data Science in detail. Discuss its characteristics, major components, and the different stages involved in the Data Science lifecycle with a suitable diagram.

77. Explain the Data Science process in detail. Discuss how Data Science is used to collect, prepare, explore, analyze, model, evaluate, and communicate insights from data with a suitable real-world example.

78. Explain the Data Science Process in detail. Discuss the role of prior knowledge, data preparation, modelling, and application with a suitable example.

79. Describe how a real-world problem can be solved using the Data Science Process. Explain the importance of prior knowledge, preparing the data, developing a model, and applying the results with a suitable example.

80. Explain datasets in detail. Discuss their characteristics, structure, and importance in Data Science with suitable examples.

81. Describe the different types of datasets used in Data Science. Explain their characteristics, features, and applications with suitable examples.

82. Explain Univariate Data and Multivariate Data in detail. Discuss their characteristics, differences, and suitable examples.

83. Describe the descriptive statistical methods used for univariate and multivariate data. Explain how these methods help in understanding and interpreting datasets.

84. Define Data Visualization and explain its importance in Data Science. Discuss how visualization helps in understanding patterns, trends, relationships, and outliers in data with suitable examples.

85. Explain the main objectives and benefits of Data Visualization in understanding and communicating data.

MODULE 2 — MACHINE LEARNING FUNDAMENTALS AND CLASSIFICATION

16. Define Machine Learning. Explain how machines learn from data with a suitable example.

17. Explain the concepts of Abstraction, Generalisation, and Evaluation in Machine Learning. Discuss how these concepts help a machine learning system learn effectively and make predictions on new data.

18. Explain the basic concept of Machine Learning and distinguish it from traditional programming.

19. Explain Supervised Learning and Unsupervised Learning with suitable examples.

20. Differentiate between Semi-Supervised Learning and Reinforcement Learning with suitable examples.

21. Explain the training and testing concept in Machine Learning. Why are training and testing data separated?

22. What is Feature Selection? Explain its importance in Machine Learning.

23. What are relevant features? Explain the importance of selecting relevant features for building an effective Machine Learning model.

24. Explain the different techniques used for Feature Selection in Machine Learning.

25. Using KNN with K = 3 and Euclidean distance, predict the result of a student with Study Hours = 4 and Attendance = 70%

26. Using KNN with K = 3, classify the new point Q(3, 3). Use Euclidean distance.

27. Using KNN with K = 5 and Euclidean distance, classify a new customer with Age = 28 and Income = 36.

28. Using the Naive Bayes Classifier, predict the result for a student with: Study = High, Attendance = High, Assignment = Yes Calculate the required prior and conditional probabilities and determine whether the student will Pass or Fail.

29. Using the Naive Bayes Classifier, predict whether a loan will be Approved or Rejected for a person with: Income = High, Credit = Good, Employment = Stable Show the probability calculations

30. Using the Naive Bayes Classifier, predict whether the person will Play = Yes or No for: Outlook = Sunny, Temperature = Cool, Wind = Weak Calculate the prior probability, conditional probabilities, and final class probabilities.

86. Explain Machine Learning in detail. Describe how machines learn from data and discuss the roles in the learning process.

87. Discuss the major concepts involved in Machine Learning and explain how a machine learning system uses data to learn patterns, generalise from examples, and evaluate its performance.

88. Explain the types of Machine Learning algorithms with suitable examples.

89. Compare Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning based on training data, learning approach, goal, output, human labeling requirements, and applications. Give suitable examples for each type.

90. Explain Feature Selection in detail. Discuss its importance, advantages, and the major techniques used for selecting relevant features in Machine Learning.

91. Explain the process of Selecting Relevant Features in Machine Learning. Discuss different Feature Selection techniques and how they help improve model performance.

92. A new fruit has Weight = 5 and Sweetness = 6. Using KNN with K = 3: a) Calculate the Euclidean distance between the new fruit and each training instance. b) Arrange the distances in ascending order. c) Identify the 3 nearest neighbours. d) Determine the class of the new fruit using majority voting. e) Explain the classification result.

93. For a new patient with Age = 42 and Blood Pressure = 142, use KNN with K = 5. a) Calculate the Euclidean distance from the new patient to every training instance. b) Identify the 5 nearest neighbours. c) Count the number of Normal and High classifications among the neighbours. d) Predict the class of the new patient using majority voting. e) State the role of K in the KNN algorithm.

94. Using the Naive Bayes Classifier with Laplace estimation, classify the following new instance: Outlook = Sunny, Temperature = Cool Calculate: a) Prior probabilities of each class. b) Conditional probabilities using Laplace estimation. c) Probability for each class. d) Compare the probabilities. e) Predict whether Play = Yes or No.

95. Using the Naive Bayes Classifier with Laplace estimation, predict whether a customer with: Age = Young, Income = Medium, Student = Yes will Purchase = Yes or No. Show the following steps: a) Calculate the prior probabilities. b) Calculate the conditional probabilities. c) Give the final classification.

MODULE 3 — DECISION TREES, CLASSIFICATION RULES AND REGRESSION

31. Explain the concept of Decision Tree Learning. Describe the main components of a decision tree with a suitable example.

32. Explain how the divide-and-conquer approach is used in decision tree learning.

33. Differentiate between the root node, internal (decision) nodes, branches, and leaf nodes of a Decision Tree with suitable examples.

34. Explain the C5.0 Decision Tree Algorithm and state its important features.

35. What is pruning in a decision tree? Explain the need for pruning and how it helps to improve the performance of a decision tree.

36. Explain the major steps involved in the construction of a decision tree using the C5.0 algorithm.

37. Explain the concept of Classification Rules with a suitable example.

38. Explain the Separate and Conquer approach used for learning classification rules.

39. Explain the 1R (One Rule) algorithm and describe its basic working.

40. Differentiate between Decision Tree Learning and Classification Rule Learning.

41. Explain how each root-to-leaf path in a Decision Tree can be represented as a classification rule.

42. Explain the concept of extracting classification rules from a Decision Tree with a suitable example.

43. Explain the concept of Regression. How is regression different from classification?

44. Explain Simple Linear Regression and describe the role of the dependent and independent variables.

45. Explain Ordinary Least Squares (OLS) estimation. How does OLS determine the best-fitting regression line?

96. Consider the following dataset for predicting whether a student will Pass or Fail based on Attendance and Internal Marks. Construct a suitable decision tree using the given data and classify a student with Attendance = Medium and Internal Mark = High. Clearly show the root node, branches, and leaf nodes.

97. The following dataset is used to predict whether a customer will Buy or Not Buy a product. Apply the divide-and-conquer approach to construct a decision tree. Show the splitting process and predict the outcome for a customer with Age = Young and Income = Low.

98. What is pruning in a decision tree? Explain the need for pruning and describe the pruning process in the C5.0 algorithm. Discuss how pruning helps to reduce overfitting and improve the generalization of the model.

99. Explain the C5.0 Decision Tree Algorithm in detail. Describe its working, the steps involved in decision tree construction, and the criteria used for selecting attributes.

100. Explain the Separate and Conquer approach for classification rule learning in detail. Describe its working, major steps, and how classification rules are generated from a dataset with a suitable example.

101. Explain the 1R algorithm in detail. Discuss the steps involved in constructing a one-rule classifier, selection of the best attribute, and classification of instances. Illustrate the algorithm with a suitable example.

102. Explain in detail how a Decision Tree can be converted into a set of classification rules. Illustrate the conversion process with a suitable Decision Tree and corresponding IF–THEN rules.

103. Compare Decision Tree Learning and Classification Rule Learning. Discuss their structure, representation, learning approach, interpretability, advantages, and limitations with suitable examples.

104. The following data represents the hours studied (X) and marks obtained (Y) by six students: Using the Ordinary Least Squares (OLS) method, calculate the regression equation of Y on X. Estimate the marks obtained by a student who studies for 8 hours.

105. Consider the following data showing the advertising expenditure (X) and sales (Y) of a company: Calculate the Pearson correlation coefficient between advertising expenditure and sales. Interpret the strength and direction of the correlation.

MODULE 4 — NEURAL NETWORKS, SVM AND CLUSTERING

46. Explain the different components of an Artificial Neuron and describe the role of weights, bias, and activation function.

47. Explain the working of an Artificial Neuron with a suitable example.

48. Explain the concept of Neural Network Learning and state its importance in Data Science.

49. Explain any four types of activation functions used in neural networks with their characteristics.

50. Explain the role of activation functions in neural networks. How do they introduce non-linearity into the network?

51. Explain the concept of an activation function. Why is an activation function required in an artificial neuron?

52. Explain the structure and working of a Single-Layer Neural Network.

53. Differentiate between Single-Layer and Multi-Layer Neural Networks.

54. Explain the functions of Input, Hidden, and Output layers in a Multi-Layer Neural Network.

55. Explain the Backpropagation Algorithm and state its main steps.

56. Explain how error is calculated in a neural network during the backpropagation process.

57. Explain the process of weight updating in the Backpropagation Algorithm. What is the role of the learning rate?

58. Explain the concept of Support Vector Machines (SVM) and state its main characteristics.

59. Explain the concept of a Maximum Margin Hyperplane in SVM. Why is maximizing the margin important?

60. Explain linearly separable data with a suitable example. How can SVM classify such data?

61. Explain the K-Means clustering algorithm and list its major steps.

62. Explain how data points are assigned to clusters in the K-Means algorithm.

63. Explain the process of updating cluster centroids in the K-Means algorithm.

64. Explain the Elbow Method for determining the appropriate number of clusters in K-Means clustering.

65. Explain how clustering results can be evaluated. Mention any two measures used for evaluating clustering performance.

66. Explain why choosing the appropriate number of clusters is important in clustering.

106. Explain Neural Network Learning and Artificial Neurons in detail. Discuss how an artificial neuron receives inputs, applies weights and bias, performs aggregation, and produces an output through an activation function.

107. Consider an artificial neuron with inputs x1=2, x2=3, weights w1=0.4, w2=0.6, and bias b=–1. Calculate the weighted sum and determine the output using a suitable step activation function. Explain the working of the neuron.

108. Explain the concept of activation functions in detail. Discuss the different types of activation functions used in neural networks, along with their characteristics and applications.

109. Discuss the commonly used activation functions such as Binary Step, Linear, Sigmoid, Tanh, and ReLU. Explain their working, characteristics, and suitable applications.

110. Compare Single-Layer and Multi-Layer Neural Networks with respect to structure, number of layers, processing capability, learning ability, and applications.

111. Explain Neural Network Topology in detail. Describe the structure and characteristics of Single-Layer and Multi-Layer Neural Networks with suitable diagrams.

112. Describe the complete process of training a neural network using the Backpropagation Algorithm. Explain how errors are calculated and propagated backward through the network to update the weights.

113. Explain the Backpropagation Algorithm in detail. Describe its working process, including forward propagation, error calculation, backward propagation, and weight updating.

114. Explain Support Vector Machines (SVM) in detail. Discuss hyperplanes, support vectors, margin, and the concept of Maximum Margin Hyperplane in classification.

115. Explain how SVM performs classification using hyperplanes. Discuss the concept of linearly separable data and explain how the Maximum Margin Hyperplane is determined for separating two classes.

116. Explain the K-Means clustering algorithm in detail. Describe the steps involved in initialization, assigning data points to clusters, updating cluster centroids, and repeating the process until convergence.

117. Describe the complete working of K-Means clustering with a suitable example. Explain how data points are assigned to clusters and how the cluster centroids are updated in each iteration.

118. Explain how clustering results are evaluated. Discuss the important criteria and measures used to assess the quality of clusters, including cohesion and separation.

119. Discuss the process of choosing and evaluating clusters in detail. Explain why the number of clusters affects the quality of clustering and describe suitable methods for selecting and evaluating the clusters.

MODULE 5 — MODEL EVALUATION AND ENSEMBLE LEARNING

67. What is a Confusion Matrix? Explain its four components

68. Explain Precision and Recall. How do they differ in evaluating a classification model?

69. Explain ROC Curves in detail

70. Explain the concept of Cross Validation. Why is it used for evaluating machine learning models?

71. What is Bootstrap Sampling? Explain how bootstrap samples are generated from a dataset.

72. Explain K-Fold Cross Validation and describe the steps involved in the procedure.

73. Explain the concept of Ensemble Learning. Why is it used to improve model performance?

74. Explain the Bagging & Boosting technique used in ensemble learning.

75. What is a Random Forest? Explain its basic structure and working.

120. A machine-learning model is used to detect whether an email is Spam or Not Spam. Out of 500 emails, the model correctly identifies 180 spam emails and 270 non-spam emails. It incorrectly classifies 20 non-spam emails as spam and 30 spam emails as non-spam. Construct the Confusion Matrix and calculate: Accuracy, Precision, Recall (Sensitivity), Specificity, F1-Measure.

121. A machine learning model is used to predict whether a patient has a particular disease. Out of 1,000 patients, the model produces the following results: 150 patients were correctly predicted as having the disease. 700 patients were correctly predicted as not having the disease. 50 healthy patients were incorrectly predicted as having the disease. 100 patients having the disease were incorrectly predicted as healthy. Construct the confusion matrix and calculate: a) Accuracy b) Precision c) Recall (Sensitivity) d) Specificity e) F1-Measure

122. A dataset contains 100 instances and is evaluated using 5-Fold Cross Validation. a) Explain how the dataset is divided into folds. b) Show how training and validation sets are formed in each iteration. c) If the accuracies obtained in the five folds are 82%, 85%, 80%, 88%, and 85%, calculate the average cross-validation accuracy. d) Explain how the average accuracy can be used to evaluate the model.

123. Explain Cross Validation techniques used for evaluating machine learning models. Discuss K-Fold Cross Validation and Bootstrap Sampling in detail, including their procedures, advantages, limitations, and their role in comparing model performance.

124. Compare Bagging, Boosting, and Random Forests with respect to their working principles, model construction, error reduction, and applications.

125. Explain the Random Forest algorithm in detail. Describe how multiple decision trees are constructed and combined to produce the final prediction. Discuss its advantages.

126. Discuss the advantages of Ensemble Methods in machine learning. Explain how ensemble techniques improve accuracy, reduce overfitting, and increase the robustness of predictive models.